Do Now · retrieval

Lesson 055 · Choosing Data Structures for Problems

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a hash collision?

Q2

When can linked-list insertion be O(1)?

Q3

What is a BST's advantage when balanced?

Q4

An empty queue receives 4, 8 and 2 in that order. Repeatedly dequeue an item and push it onto an empty stack until the queue is empty. Trace the sequence returned by popping the stack until it is empty.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Discover — Choose structures from operations first, not familiar names: indexed access, insertion, deletion, queue discipline or key lookup.

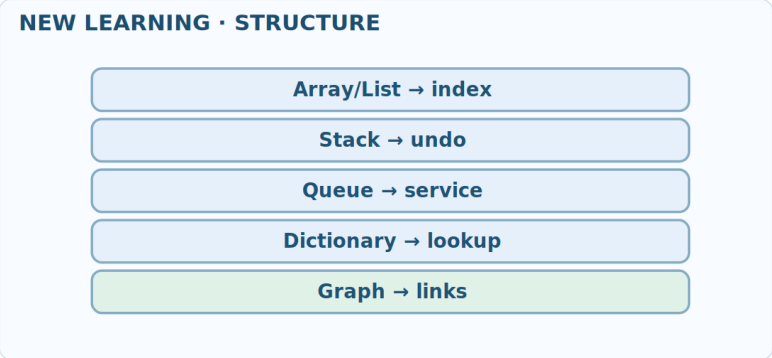
Core idea 2

Trace — Consider runtime and additional memory: arrays/lists support indexed access; linked lists can insert cheaply given a node; dictionaries use extra bucket storage.

Core idea 3

Evaluate — Justify the suitability for a real task, including constraints and the cost of maintaining the structure.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

operation — Action such as lookup or insertion.

random access — Direct access by numeric index.

trade-off — A benefit weighed against a cost.

complexity — Growth in resource use.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

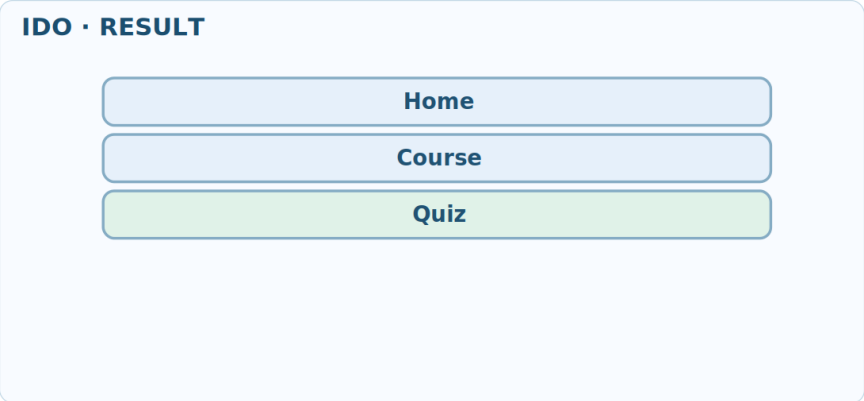
Problem

Choose a structure for a browser's Back action and justify.

Follow the worked method

1. Explain the trace — Stack (LIFO): the most recently visited earlier page is retrieved first. Browser navigation may use multiple stacks for back/forward.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the trace
Queue FIFO serves earliest arrivals first, subject to real-world priority policies.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Match an undo feature, printer queue, pupil ID lookup and road map to a structure. [4]

Question 2

A timetable stores fixed grid of rooms and periods. Recommend a 2D array or graph and explain operations. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate using List<T> versus linked list for a million-item collection, focusing on frequent random index access versus middle insertion with a known node. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The same data structure is always optimal for every collection.

Use key vocabulary from memory

1. Explain operation in your own words.

2. Explain random access in your own words.

3. Explain trade-off in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.